ITDS 124 Mathematics of Computing

Chapter 3. Counting

3.2 Combinations

การจัดหมู่

Created by : Rawesak Tanawongsuwan
rawesak.tan@mahidol.ac.th

Combinations

- The multiplication principle and the counting methods for permutations all apply to situations where order matters.

- Now, we look at some counting problems where order does not matter.

Counting problem 2

- Let A be any set with n elements and $0 \leq r \leq n$. How many different subsets of A are there, each with r elements?

Compare counting problems 1 and 2

Problem 1

How many different sequences, each of length r, can be formed using elements from A if

(a) elements in the sequence may be repeated?

(b) all elements in the sequence must be distinct?

Problem 2

Let A be any set with n elements and $0 \leq r \leq n$. How many different subsets of A are there, each with r elements?

Compare counting problem 1 and 2

Problem 1

How many different sequences, each of length r, can be formed using elements from A if

(a) elements in the sequence may be repeated?

(b) all elements in the sequence must be distinct?

Problem 2

Let A be any set with n elements and $0 \leq r \leq n$. How many different subsets of A are there, each with r elements?

Counting problem 2

- Let A be any set with n elements and $0 \leq r \leq n$. How many different subsets of A are there, each with r elements?
  - Note that an r-element subset of an n-element set A is **a combination of A, taken r at a time**.

Note that this problem is different from counting problem 1 in that the order of the elements is irrelevant.

Example 1: Let $A = \{1,2,3,4\}$. The following are all distinct combinations of A, taken 3 at a time : $A_1 = \{1,2,3\}, A_2 = \{1,2,4\}, A_3 = \{1,3,4\}, A_4 = \{2,3,4\}$. Note that these are subsets, not sequences.

Example 2: Let A be the set of all 52 cards in a deck of cards. Then a combination of A, taken five at a time, is just a hand of five cards regardless of how these cards were dealt.

Counter problem 2

- Let A be any set with n elements and $0 \leq r \leq n$. How many different subsets of A are there, each with r elements?
  - Note that an r-element subset of an n-element set A is **a combination of A, taken r at a time**.

Observe that each permutation of the elements of A, taken r at a time, can be produced by performing the following two tasks in sequence.

Task 1: Choose a subset B of A containing r elements.

Task 2: Choose a particular permutation of B.

We are trying to compute the number of ways to choose B. Call this number C.

Then task 1 can be performed in C ways, and task 2 can be performed in $r!$ ways.

Thus the total number of ways of performing both tasks is, $C * r!$. But it is also $P(n, r)$.

Hence, $C * r! = P(n, r) = \frac{n!}{(n-r)!}$ ➔ $C = \frac{n!}{r!(n-r)!}$

Theorem 1

- Let A be a set with $|A| = n$, and let $0 \leq r \leq n$. Then the number of combinations of the elements of A, taken r at a time, that is, the number of r-element subsets of A, or the number of combinations of n objects taken r at a time is

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

Dealing cards example

Compute the number of distinct five-card hands that can be dealt from a deck of 52 cards.

This number is $C(52,5) = \binom{52}{5}$ because the order in which the cards were dealt is irrelevant.

$$C(52,5) = \binom{52}{5} = \frac{52!}{5!\,47!} = 2{,}598{,}960$$

Combinations with Repetition

- So far, when we discuss combination
- **A combination of A, taken r at a time ➔ r elements are distinct.**
- Next we consider **combinations with repetition of elements allowed**.
- Given n object types, we want to select r objects with replacement or repetition is allowed.

Example of Combinations with Repetition

- In a donut shop, there are 4 types of donuts. How many ways can we select 3 piece of donuts to take home? Note that you can get the same types of donut. $n = 4$

- Why $4 * 4 * 4 = 4^3$ is not the correct answer?

Example1 of Combinations with Repetition

- In a donut shop, there are 4 types of donuts. How many ways can we select 3 piece of donuts to take home? Note that you can get the same types of donut.

Theorem 2

- Suppose k selections are to be made from n items without regard to order and repeats are allowed, assuming at least k copies of each of the n items. The number of ways these selections can be made is

$$C(n+k-1,k) = \binom{n+k-1}{k}$$

Example2 of Combinations with Repetition

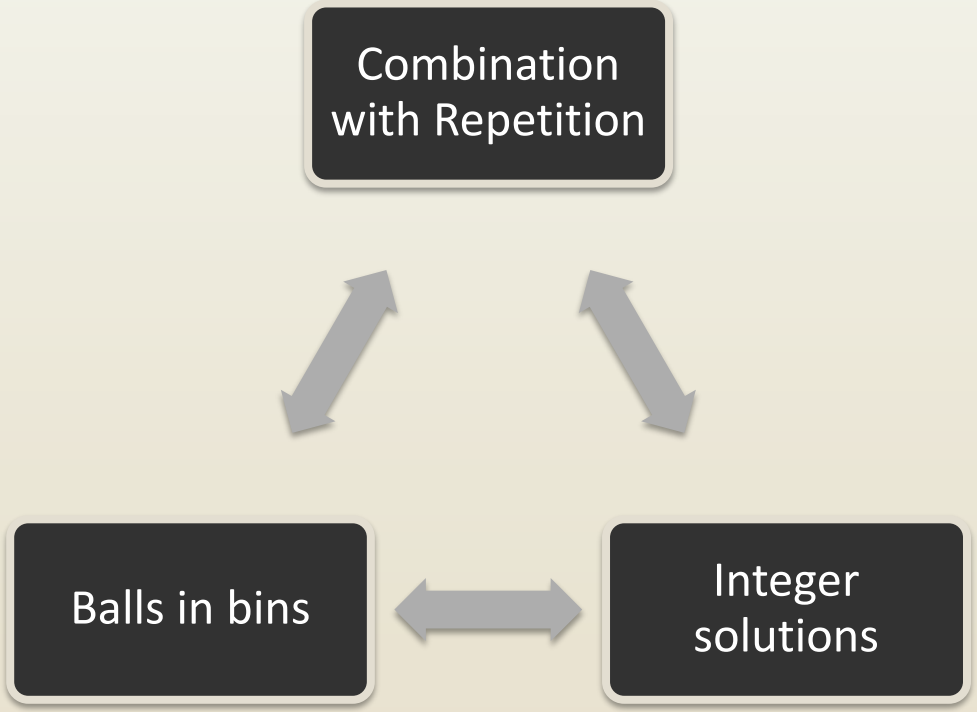
In how many ways can a person choose three ice cream scoops from the list of 10 ice cream flavors if repeats are allowed?

Example3 : Integer solutions (Balls in bins)

- Suppose we have 3 integers such that $x_1 + x_2 + x_3 = 7$, where $x_1, x_2, x_3 \geq 0$. How many integer solutions are there?

Example3 : Integer solutions (Combination with Repetition)

- Suppose we have 3 integers such that $x_1 + x_2 + x_3 = 7$, where $x_1, x_2, x_3 \geq 0$. How many integer solutions are there?



Integer solutions vs Combination with Repetition

$x_1 + x_2 + \cdots + x_n = r$, where $x_i \geq 0$.

How many integer solutions are there?

Same as

Given n object types, we want to select r objects with replacement or repetition is allowed"

Solution $\binom{n+r-1}{r}$

Example4

- $x_1 + x_2 = 1$, where $x_1, x_2 \geq 0$. How many integer solutions are there?

Example5

- $x_1 + x_2 + x_3 = 7$, where $x_1, x_2, x_3 \geq 0$. How many integer solutions are there?

Example6

- $x_1 + x_2 + \ldots + x_{20} = 12$, where $x_i \geq 0$. How many integer solutions are there?

Balls in bins vs Integer solutions

- How many ways can we put 10 identical balls into 6 different bins?

- Now turn this into integer solutions problem

Integer solutions (harder) → Books on Shelves

$x_1 + x_2 + x_3 = 9$ $\quad\quad$ $x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 > 1$

How many integer solutions are there?

Integer solutions (harder) → Variable substitution

$x_1 + x_2 + x_3 = 9$ $\quad\quad$ $x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 > 1$

How many integer solutions are there?

More examples (1)

- Example 1 : Suppose that a valid computer password consists of seven characters, the first of which is a letter chosen from the set $\{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{F}, \mathrm{G}\}$ and the remaining six characters are letters chosen from the English alphabet or a digit. How many different passwords are possible?

Password can be constructed by performing the tasks T_1 and T_2 in sequence.

Task 1: Choose a starting letter from the set given ➔ $C(7,1) = 7$

Task 2: Choose a sequence of letter and digits. Repeats are allowed. ➔ $36^6 = 2{,}176{,}782{,}336$

By the multiplication principle, there are $7 * 2{,}176{,}782{,}336 = 15{,}237{,}476{,}352$ different passwords.

More examples (2)

- Example 2 : How many different seven-person committees can be formed each containing three women from an available set of 20 women and four men from an available set of 30 men?

In this case a committee can be formed by performing the following two tasks in succession:

Task 1: Choose three women from the set of 20 women ➔ $C(20,3) = 1{,}140$

Task 2: Choose four men from the set of 30 men ➔ $C(30,4) = 27{,}405$

By the multiplication principle, there are $1{,}140 * 27{,}405 = 31{,}241{,}700$ different committees.